



BICING PREDICTIONS 2024

DIEDERIK DE VRIES | SIDDHARTH | SHARON | NUPUR

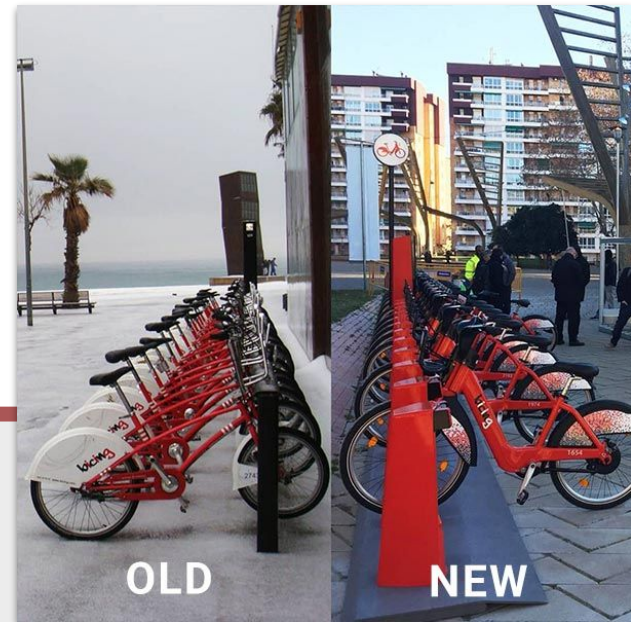
Source: www.unsplash.com

WHAT IS BICING?

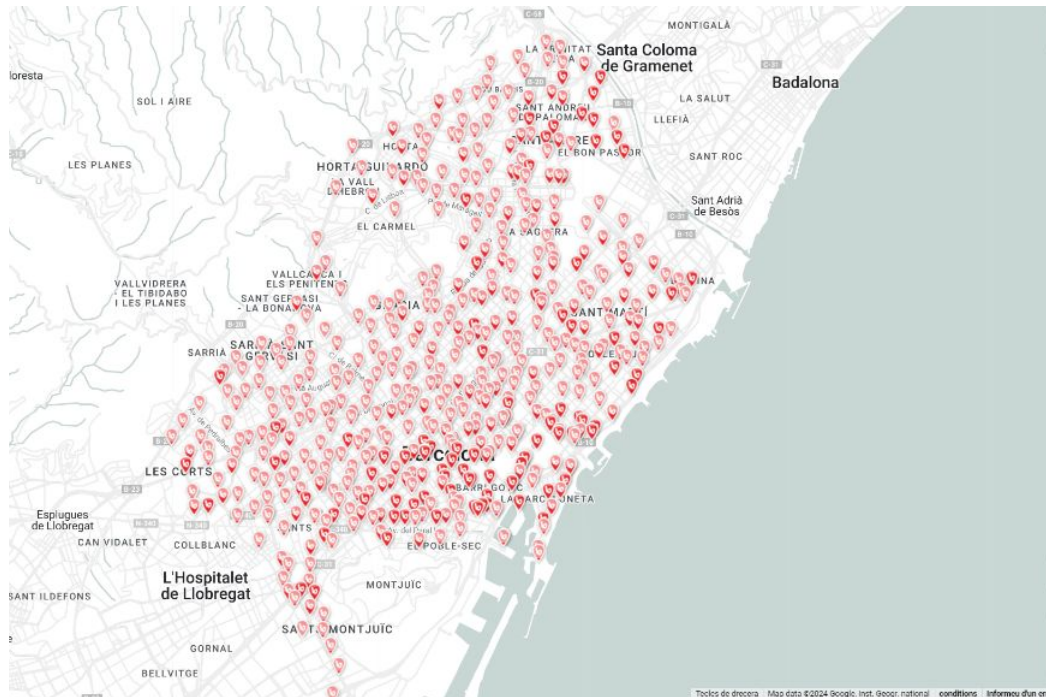
Bicing was launched in **2007** as a viable eco-friendly alternative or complement to traditional private and public transportation.

The mobility network has also allowed for increased physical activity and better connectivity between neighbourhoods of Barcelona.

It is aimed at anyone with a NIE over 16 years of age... (and 50€, annually).
The **bicing 2.0** models were released to include electrical charging docks.

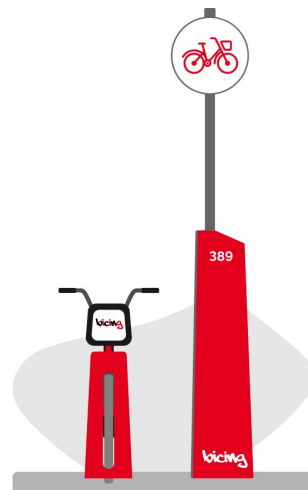


WHAT IS BICING?



Total Number of Bikes by end of 2024 : **8,000**
Electric Bikes : **5,000**

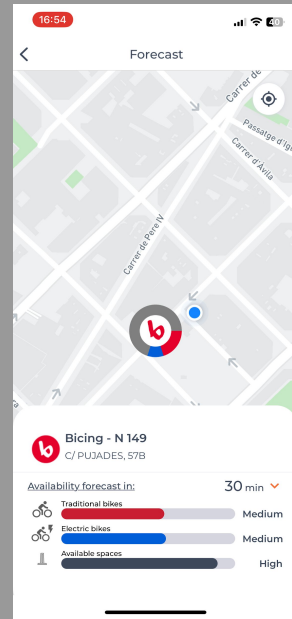
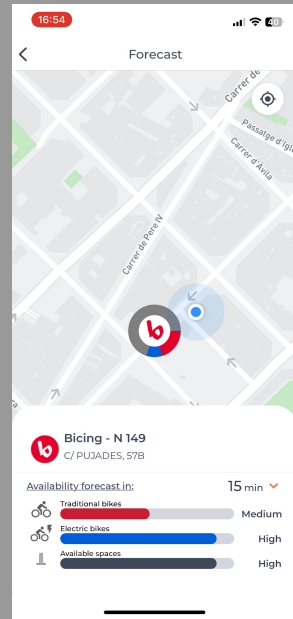
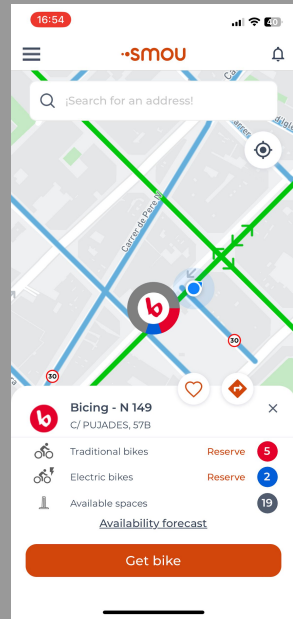
On a working day, Bicing gets over
60,000 users



The city of Barcelona has a network of bike lanes of
more than 200km

CURRENT FORECASTING METHOD

16:54



15 minutes
30 minutes

- Traditional
- Electric
- Spaces

LOW
MEDIUM
HIGH

TASK : “The Second Traditional Great Machine Learning Race”

What was the availability of biking bikes at any given station in Barcelona at any given time during the first three months of 2024?

STATION DATA

station_id		name	physical_configuration	lat	lon	altitude	address	post_code	capacity	is_charging_station	nearby_distance	x_ride_code_support	rental_uris	cross_street	last_updated	tll	V1	is_valet_station	x_valet_station_details
0	1.0	GRAN VIA CORTS CATALANES, 760	ELECTRICBIKESTATION	41.397978	2.180107	16.0	GRAN VIA CORTS CATALANES, 760	8013.0	46.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
1	2.0	C/ ROGER DE FLOR, 126	ELECTRICBIKESTATION	41.395488	2.177198	17.0	C/ ROGER DE FLOR, 126	8013.0	29.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
2	3.0	C/ NAPOLS, 82	ELECTRICBIKESTATION	41.394156	2.181331	11.0	C/ NAPOLS, 82	8013.0	27.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
3	4.0	C/ RIBES, 13	ELECTRICBIKESTATION	41.393317	2.181248	8.0	C/ RIBES, 13	8013.0	21.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
4	5.0	PG. LLUIS COMPANYNS, 11 (ARC TRIOMF)	ELECTRICBIKESTATION	41.391103	2.180176	7.0	PG. LLUIS COMPANYNS, 11 (ARC TRIOMF)	8018.0	39.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
506	516.0	C/ SANT ADRIA, 88	ELECTRICBIKESTATION	41.435460	2.200157	15.0	C/ SANT ADRIA, 88	8030.0	21.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
507	517.0	AV. RASOS DE PEGUERA, 10	ELECTRICBIKESTATION	41.462095	2.178959	44.0	AV. RASOS DE PEGUERA, 10	8033.0	20.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
508	518.0	C/ LLOBREGÓS, 115	ELECTRICBIKESTATION	41.424689	2.157049	112.0	C/ LLOBREGÓS, 115	8032.0	27.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
509	519.0	C/ PEDRELL, 52	ELECTRICBIKESTATION	41.424655	2.166289	110.0	C/ PEDRELL, 52	8032.0	24.0	True	1000.0	True	NaN	NaN	1.709248e+09	0.0	NaN	NaN	NaN
510	535.0	Gera_Testing - 535	VALET	41.347594	2.119498	NaN	Gera_Testing - 535	NaN	1.0	False	1.0	True	NaN	NaN	1.710170e+09	0.0	NaN	True	NaN

Source: [bicing_stations.csv](#)

STATION DATA AVAILABILITY 2019-2023

station_id	year	month	day	hour	num_bikes_available_types.mechanical	num_bikes_available_types.ebike	availability	
0	1	2023	1	1	0	32.066667	2.666667	0.239130
1	1	2023	1	1	1	36.666667	2.666667	0.144928
2	1	2023	1	1	2	37.416667	1.250000	0.157609
3	1	2023	1	1	3	37.833333	1.750000	0.135870
4	1	2023	1	1	4	38.250000	1.083333	0.143116
...	...	...	...	...	...	...	...	
4110207	519	2023	12	31	19	0.000000	2.166667	0.868056
4110208	519	2023	12	31	20	0.000000	4.583333	0.767361
4110209	519	2023	12	31	21	0.000000	5.833333	0.715278
4110210	519	2023	12	31	22	0.000000	6.583333	0.684028
4110211	519	2023	12	31	23	0.000000	7.000000	0.666667

Source: *bicing_vacancy_2023.csv*

BIKE SHARING: A Growing Trend in Barcelona

2007

Bicing implemented in Barcelona by the City Council

March

Difficulty in finding spots to park or pick up bikes amongst people

November

2019

Started showing irregular patterns in the bike availability due to covid

2021

The availability of bikes on the bike stands are more graduated over the years

2023

Train the model between 2021- 2023

2024

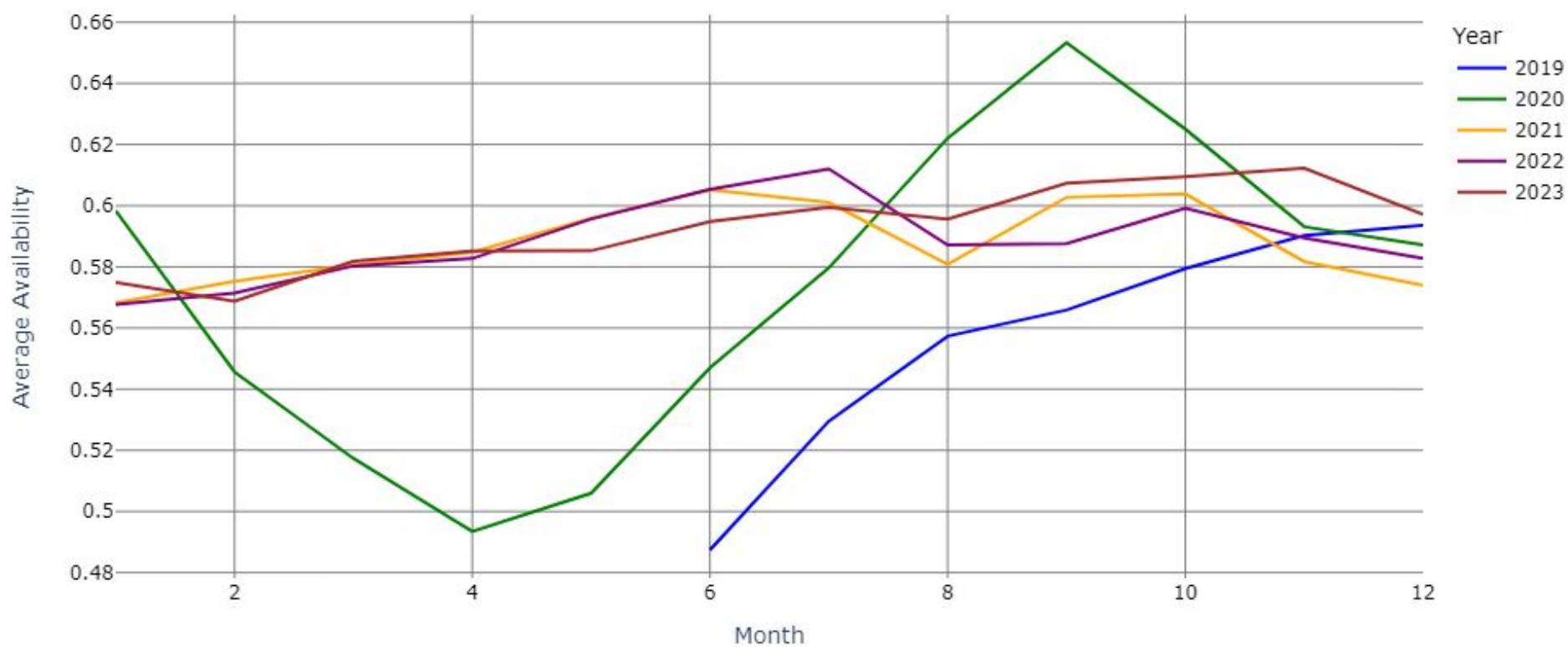
Train the model to predict bike availability for 2024

2025

509stations

AS OF 2023 IN BARCELONA

Average Bicing Vacancy Over Months (2019-2023)



FACTORS THAT AFFECT THE PREDICTION

WEATHER

Source: www.unsplash.com

NATURE OF THE DAY

POPULATION

The Meteostat python library allowed to access hourly data for the city of Barcelona.

Since weather is cyclical, it is thereby an important factor in the prediction of Bicing



Meteostat Developers

Meteostat is an open platform which provides free access to historical weather and climate data.

💡 Click here to ask Blackbox to help you code faster

```
from datetime import datetime
import matplotlib.pyplot as plt
from meteostat import Point, Daily
```

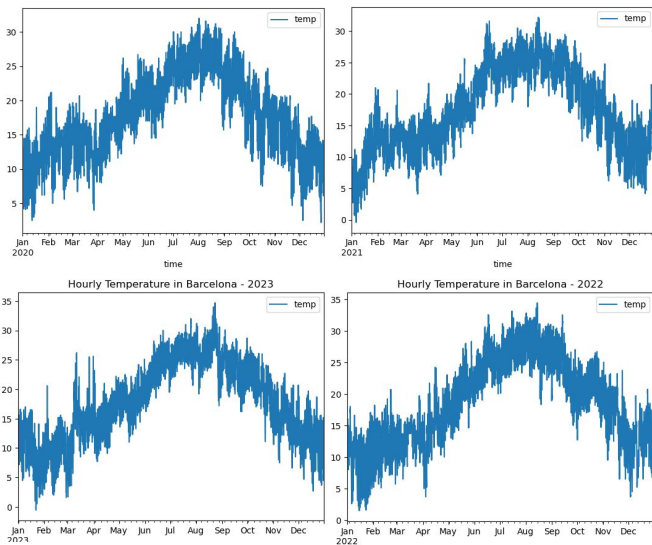
```
# Set time period
start = datetime(2019, 1, 1)
end = datetime(2019, 12, 31)
```

```
# Create Point for Barcelona, BC
barcelona = Point(41.3828939, 2.1774322, 12)
```

```
# Get daily data for 2018
data = Daily(barcelona, start, end)
data = data.fetch()
```

```
# Plot line chart including average, minimum and maximum temperature
data.plot(y=['tavg', 'tmin', 'tmax'])
plt.show()
```

WEATHER



CYCLICAL PATTERNS

	ID	temp	dwpt	rhum	prcp	snow	wdir	wspd	wpgt	pres	tsun	coco
0	2024010100	7.5	3.4	75.0	0.0	NaN	10.0	5.4	16.7	1017.8	NaN	4.0
1	2024010101	8.0	3.6	74.0	0.0	NaN	350.0	14.8	18.5	1018.2	NaN	2.0
2	2024010102	7.7	3.7	76.0	0.0	NaN	350.0	14.8	20.4	1018.1	NaN	2.0
3	2024010103	7.5	3.5	76.0	0.0	NaN	350.0	14.8	18.5	1018.0	NaN	2.0
4	2024010104	7.7	3.4	74.0	0.0	NaN	360.0	13.0	20.4	1017.7	NaN	2.0

	ID	temp	rhum
0	2024010100	7.5	75.0
1	2024010101	8.0	74.0
2	2024010102	7.7	76.0
3	2024010103	7.5	76.0
4	2024010104	7.7	74.0

SORTING FOR RELEVANT DATA

NATURE OF THE DAY

JANUARY						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

FEBRUARY						
M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

MARCH						
M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

APRIL						
M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

MAY						
M	T	W	T	F	S	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

JUNE						
M	T	W	T	F	S	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

JULY						
M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

AUGUST						
M	T	W	T	F	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

SEPTEMBER						
M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

OCTOBER						
M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

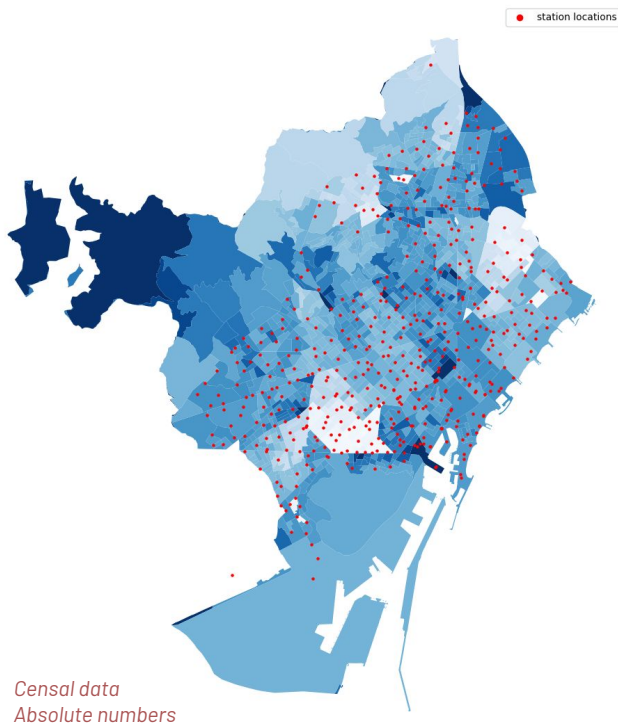
NOVEMBER						
M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

DECEMBER						
M	T	W	T	F	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Assigning a **boolean value** to all the days over the year, where all working days of the year are assigned a value of 0 and a value of 1 to all weekends and public holidays in the year.

POPULATION

The population and population density of the censal sections would affect the availability of bikes in the station within the section.



POPULATION

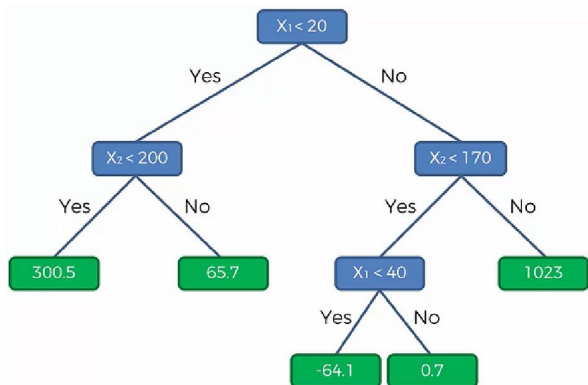
	MUNICIPI	DISTRICTE	SECCIO	MUNDISSEC	YEAR	SEC	AGE	NATION	SEX	CONCEPT	status	value
0	080193	01	001	08019301001	2021	8019301001	Y000_015	ESPA	M	POP	p	30
1	080193	01	001	08019301001	2021	8019301001	Y000_015	ESPA	F	POP	p	39
2	080193	01	001	08019301001	2021	8019301001	Y000_015	ESTR	M	POP	p	27
3	080193	01	001	08019301001	2021	8019301001	Y000_015	ESTR	F	POP	p	48
4	080193	01	001	08019301001	2021	8019301001	Y016_064	ESPA	M	POP	p	258
...	...	...	...	...	...	...	...	...	...	...	...	...
13327	080193	10	143	08019310143	2021	8019310143	Y016_064	ESTR	F	POP	p	120
13328	080193	10	143	08019310143	2021	8019310143	Y_GE065	ESPA	M	POP	p	147
13329	080193	10	143	08019310143	2021	8019310143	Y_GE065	ESPA	F	POP	p	234
13330	080193	10	143	08019310143	2021	8019310143	Y_GE065	ESTR	M	POP	p	3
13331	080193	10	143	08019310143	2021	8019310143	Y_GE065	ESTR	F	POP	p	6

Taking sum of the gender because that could give duplicate values when concatenated with the data preparation dataframe



RUNNING MULTIPLE MODELS

Decision Tree Regressor



MSE train: 0.064

Train model

```
In [2]: DF = pd.read_feather('training_data.feather').set_index('ID')

X = DF.drop(columns=['availability']) # Features
y = DF['availability'] # Target variable

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
In [3]: from sklearn.tree import DecisionTreeRegressor

# Create Decision Tree regressor object
clf = DecisionTreeRegressor()

# Train Decision Tree regressor
clf.fit(X_train, y_train)
```

```
Out[3]: DecisionTreeRegressor()

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.
```

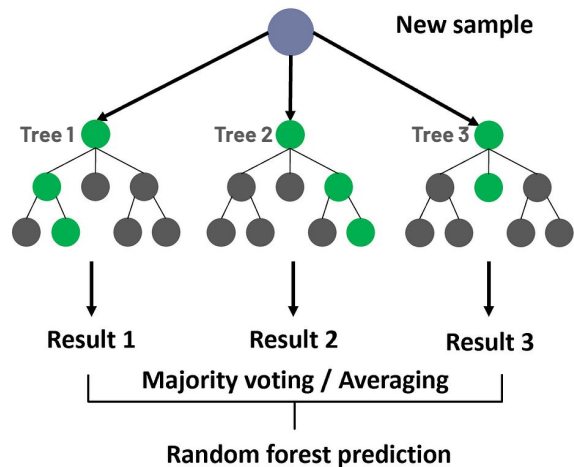
```
In [4]: from sklearn.metrics import mean_squared_error

# Predict target variable for testing data
y_pred = clf.predict(X_test)

# Calculate Mean Squared Error (MSE)
mse = mean_squared_error(y_test, y_pred)
print("Mean Squared Error (MSE):", mse)
```

Mean Squared Error (MSE): 0.0637562163843361

Random Forest Regressor



MSE train: 0.023

Train model

```
In [18]: DF = pd.read_feather('training_data.feather').set_index('ID')
```

```
In [19]: #Missing values
df_rfr_num = DF[['rhum']].copy()

#Missing values
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
imputer.fit(df_rfr_num)
df_rfr_imputer = imputer.transform(df_rfr_num)
df_rfr_num = pd.DataFrame(data=df_rfr_imputer, columns=df_rfr_num.columns, index=df_rfr_num.index)
DF['rhum'] = df_rfr_num['rhum']
```

```
In [47]: X = DF.drop(columns=['availability']) # Features
y = DF['availability'] # Target variable

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
In [ ]: #DONT RUNNNNN

rf_reg = RandomForestRegressor(n_estimators=10)

rf_reg.fit(X_train, y_train)
```

```
In [52]: y_pred = rf_reg.predict(X_test)
```

```
In [54]: y_pred.shape
```

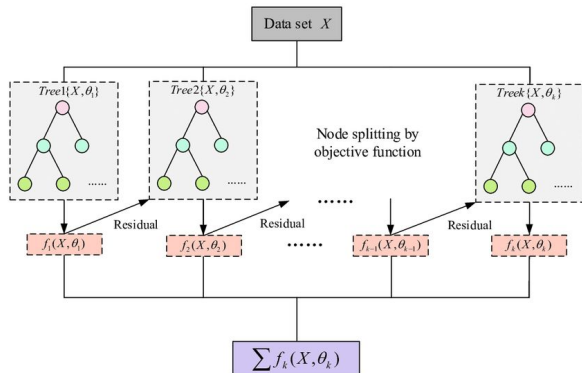
```
Out[54]: (2588837,)
```

```
In [53]: from sklearn.metrics import mean_squared_error
```

```
# Calculate Mean Squared Error (MSE)
mse = mean_squared_error(y_test, y_pred)
print("Mean Squared Error (MSE):", mse)
```

Mean Squared Error (MSE): 0.02262132361839028

XG Booster Regressor



MSE train: 0.057

```
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.ensemble import GradientBoostingRegressor

# Create a pipeline that includes an imputer transformer
pipeline = Pipeline([
    ('imputer', SimpleImputer(strategy='mean')),
    ('gbm', GradientBoostingRegressor())
])

# Train the pipeline on the training data
pipeline.fit(X_train, y_train)

# Make predictions on the testing data
y_pred = pipeline.predict(X_test)
```

```
# Print the predicted values
print("Predicted values: ", y_pred)
```

Predicted values: [0.6105014 0.59217103 0.4397086 ... 0.58747683 0.5790057 0.50688578]

```
from sklearn.metrics import mean_squared_error
```

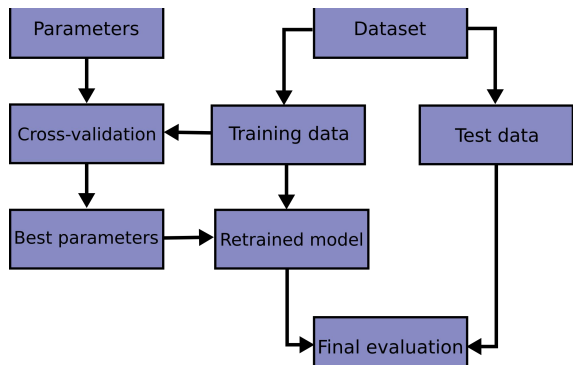
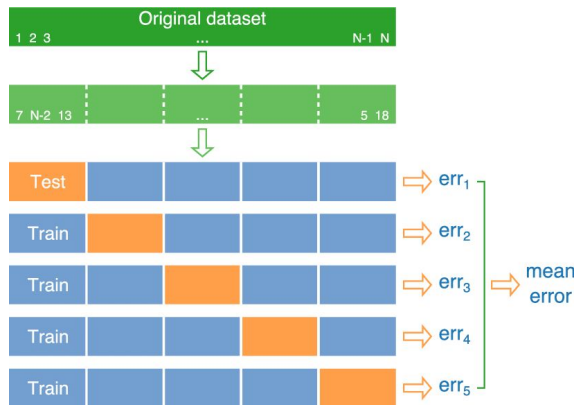
```
# Make predictions on the testing data
y_pred = pipeline.predict(X_test)
```

```
# Calculate the mean squared error
mse = mean_squared_error(y_test, y_pred)
```

```
print("Mean Squared Error: ", mse)
```

Mean Squared Error: 0.05725079084150724

Cross Validation



```
from sklearn.model_selection import cross_val_score

# Perform cross-validation
scores = cross_val_score(pipeline, X, y, cv=5)

# Print the mean and standard deviation of the scores
print("Mean score: ", scores.mean())
print("Standard deviation: ", scores.std())
```

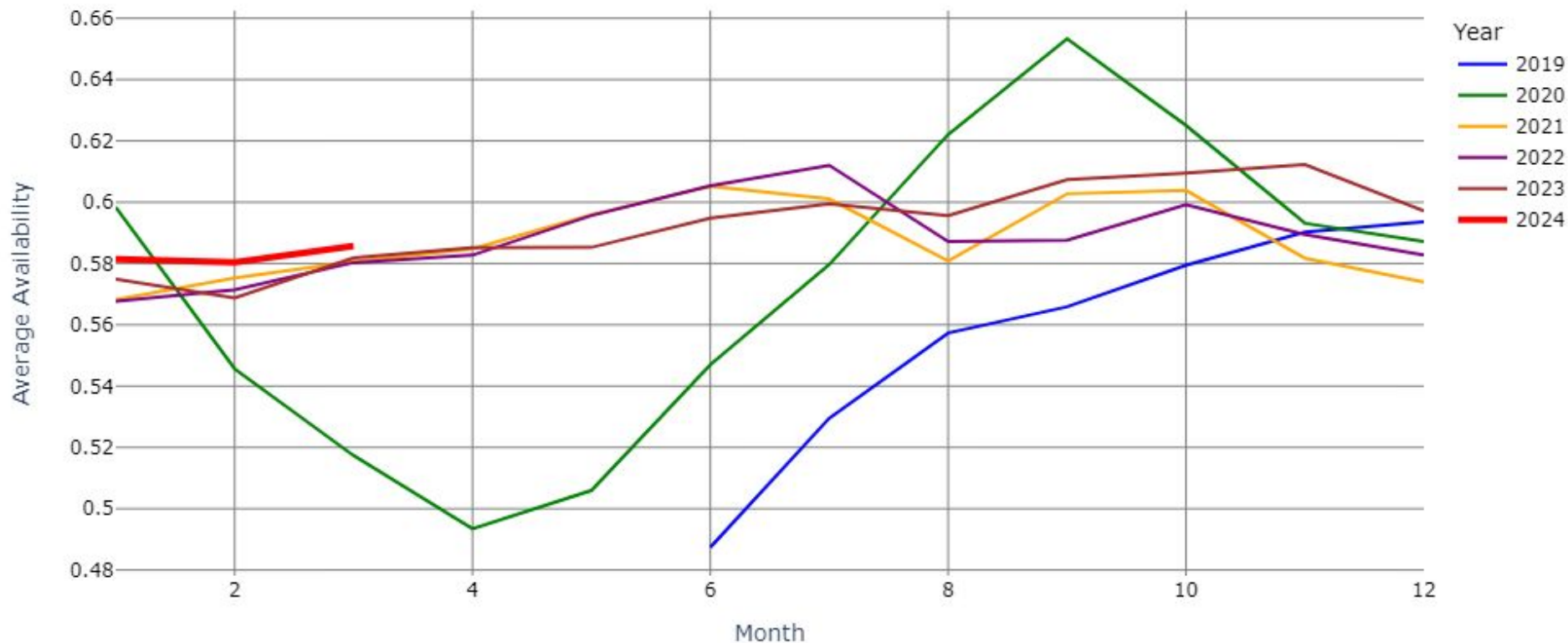
Mean score: 0.20179331157694264
Standard deviation: 0.04123203425293613

What do the scores mean?

In your case, a mean score of 0.20179331157694264 suggests that the model performs reasonably well, with a relatively low error rate.

The standard deviation of 0.04123203425293613 indicates that there is some variability in the model's performance, but it is not too high, suggesting that the model's performance is relatively consistent.

Average Bicing Vacancy Over Months (2019-2024)





Source: www.unsplash.com

THANK YOU

DIEDERIK DE VRIES | SIDDHARTH | SHARON | NUPUR